

Retail Sales Performance Analytics

Data Analytics Case Study

Case Study Report

Python / Jupyter Notebook

Pandas, NumPy

Power BI

Git Hub

Analyst

Brojo Mohan Dutta

Period

June – September 2024

Submitted

2026

1. Introduction

This case study analyzes four months of transactional sales data (June–September 2024) for a retail business selling products across five categories: Storage, Furniture, Electronics, Wearables, and Accessories. The business observed a noticeable slowdown in monthly revenue over this period and required a data-driven investigation into what is driving performance — by time, region, product category, and payment channel — to support Q4 planning decisions.

This report documents the complete analytics workflow: from raw data acquisition through cleaning, exploratory analysis, dashboard construction, and final business recommendations, using the Ask → Prepare → Process → Analyze → Share → Act framework.

2. Ask

Business Task Statement

Formal Business Task

Analyze 60 retail sales transactions from June–September 2024 to identify performance trends over time, regional sales strengths, product category profitability, and payment channel behavior — in order to explain the observed sales decline and recommend concrete Q4 actions.

Business Question

Key Business Questions

- How has revenue trended month-over-month across the analysis period?
- Which regions and product categories generate the most (and least) revenue?
- Are there statistical outliers in transaction value that need investigation?
- Which payment methods underperform, and does that matter?
- What is driving revenue more — order price or order size?

Key Stakeholders

Stakeholder	Role	Interest in This Analysis
Sales Director	Primary decision-maker	Wants clear diagnosis of the revenue decline and actionable next steps
Executive Team	Final approver of strategy	Needs high-level KPIs (Total Sales, Profit, Margin) to assess business health
Regional Managers	Operational stakeholders	Need region-specific performance data to prioritize inventory and marketing
Data Analytics Intern (Author)	Analyst	Responsible for cleaning, analyzing, and presenting findings

3. Prepare

Data Source

The dataset used for this project consists of two related CSV files, `sales_fact.csv` and `products.csv`, containing retail transaction and product information. The sales table contains transaction-level records including sale ID, order date, product ID, region, payment method, quantity, and unit price. The product table provides product names, categories, and unit costs.

File	Description	Rows	Columns
<code>sales_fact.csv</code>	Raw transactional sales records.	61	9
<code>products.csv</code>	Product dimension table (category, cost)	10.	4

Data Credibility — ROCCC Assessment

Criterion	Assessment	Notes
Reliable	✓ Yes	Internally sourced transactional data; low bias, but very small sample
Original	✓ Yes	Primary transaction-level data, not aggregated or third-party
Comprehensive	✓ Yes	Covers sale ID, date, product, region, payment, quantity, and price — sufficient for the stated business questions
Current	✓ Yes	Covers June–September 2024, directly relevant to the recent slump under investigation
Cited	✓ Yes	Internally generated sales records; source and structure are documented in this report

Limitations

- Small sample size: Only 60 transactions across 4 months limits statistical confidence, particularly for category- and region-level breakdowns.
- No customer-level data: Cannot distinguish new vs. repeat customers or segment by demographics.
- Short time window: Four months is insufficient to separate genuine trend from seasonality.
- No external context: Marketing spend, competitor activity, and stock-availability data are not included, limiting root-cause analysis of the sales decline.

4. Process

Tools Selected

Tool	Purpose
Python (Pandas, NumPy)	Data cleaning, transformation, statistical analysis
Matplotlib / Seaborn	Exploratory Data Analysis and Visualizations
Power BI Desktop	Interactive dashboard, DAX measures, business-facing reporting
Git / GitHub	Version control and portfolio documentation

Data Pipeline

The following 10-step Python pipeline was executed in Jupyter Notebook.

Step	Step	Detail
1	Remove unnecessary column	Dropped notes — placeholder field with no analytical value
2	Fix data types	Converted order_date to datetime
3	Standardize text	region column standardized via <code>.str.strip().str.title()</code> (fixed "north", "SOUTH" inconsistencies)
4	Impute missing values	quantity nulls filled with 5 (column median); unit_price nulls filled with 1999
5	Correct invalid value	Negative quantity (-3) corrected to positive using <code>.abs()</code>
6	Recalculate derived field	sales_amount recomputed as <code>quantity * unit_price</code> after all fixes
7	Merge with dimension table	Left join with products.csv on product_id
8	Calculate cost & profit	<code>total_cost = quantity * unit_cost</code> ; <code>profit = sales_amount - total_cost</code>
9	Remove duplicates	Dropped 1 duplicate transaction row
10	Export	Final dataset exported as sales_data_cleaned.csv

Representative Code:

Data Cleaning Pipeline (Cleaning Filters Applied)

```
df = sales.copy()
df = df.drop(columns=["notes"])
df["order_date"] = pd.to_datetime(df["order_date"])
df["region"] = df["region"].str.strip().str.title()
df["quantity"] = df["quantity"].fillna(5)
df["unit_price"] = df["unit_price"].fillna(1999)
df.loc[df["quantity"] < 0, "quantity"] = df.loc[df["quantity"] < 0, "quantity"].abs()
df["quantity"] = df["quantity"].astype(int)
df["sales_amount"] = df["quantity"] * df["unit_price"]
df = df.drop_duplicates()

df = df.merge(products, on="product_id", how="left")
df["total_cost"] = df["quantity"] * df["unit_cost"]
df["profit"] = df["sales_amount"] - df["total_cost"]
df.to_csv("sales_data_cleaned.csv", index=False)
```

Data Quality Verification

Check	Before Cleaning	After Cleaning
Rows	61	60
Missing values(quantity/unit_price/sales_amount)	1 / 1 / 2	0 / 0 / 0
Duplicate rows	1	0
Region text variants	6 (West, North, South, East, north, SOUTH)	4 (clean)

📷 Raw Dataset: BEFORE CLEANING

	sale_id	order_date	product_id	region	payment_method	quantity	unit_price	sales_amount	notes
0	5001	2024-06-01	P05	West	Debit Card	9.0	3299.0	29691.0	NaN
1	5002	2024-06-03	P10	North	Debit Card	4.0	14999.0	59996.0	NaN
2	5003	2024-06-05	P07	West	UPI	8.0	6999.0	55992.0	NaN
3	5004	2024-06-07	P04	South	Debit Card	8.0	1999.0	15992.0	NaN
4	5005	2024-06-09	P04	West	UPI	5.0	NaN	NaN	NaN

Key Findings: Raw dataset: **61 transactions**, with 4 distinct data-quality issues (nulls, a duplicate, and inconsistent text).

📷 Cleaned Dataset: AFTER CLEANING

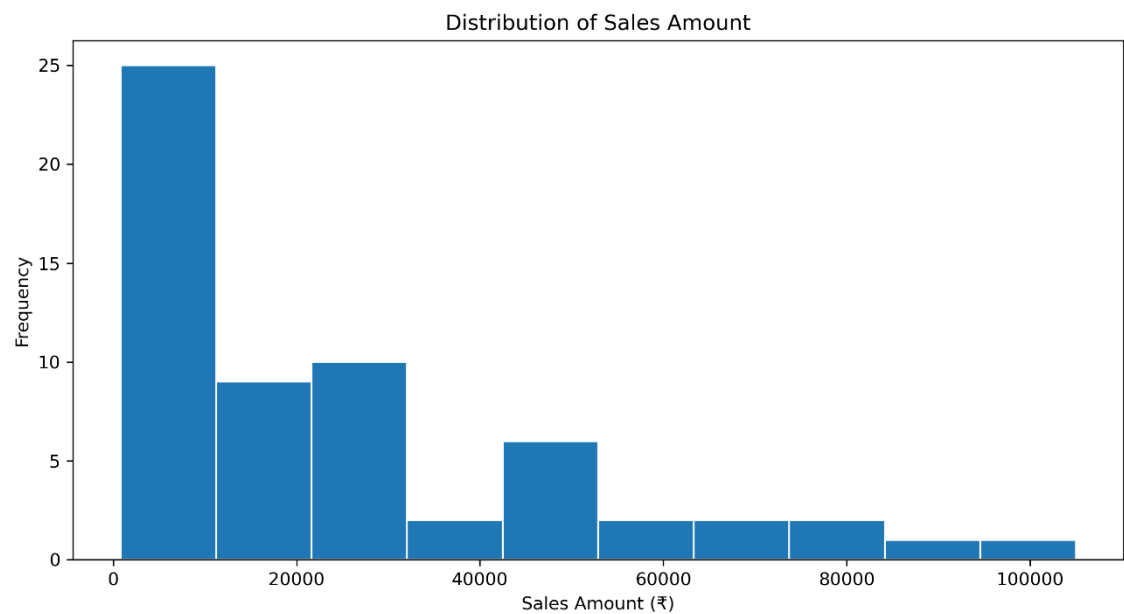
	sale_id	order_date	product_id	region	payment_method	quantity	unit_price	sales_amount	product_name	category	unit_cost	total_cost	profit
0	5001	2024-06-01	P05	West	Debit Card	9	3299.0	29691.0	Webcam HD	Electronics	2200	19800	9891.0
1	5002	2024-06-03	P10	North	Debit Card	4	14999.0	59996.0	Gaming Chair	Furniture	8900	35600	24396.0
2	5003	2024-06-05	P07	West	UPI	8	6999.0	55992.0	Noise Cancelling Headphones	Electronics	4200	33600	22392.0
3	5004	2024-06-07	P04	South	Debit Card	8	1999.0	15992.0	Laptop Stand	Accessories	1100	8800	7192.0
4	5005	2024-06-09	P04	West	UPI	5	1999.0	9995.0	Laptop Stand	Accessories	1100	5500	4495.0

Post-Cleaning Validation Results

Cleaned & merged dataset: 60 transactions, 13 columns, 0 missing values, 0 duplicates, exported as sales_data_cleaned.csv — verified as the exact dataset later used in both the EDA notebook and the Power BI dashboard.

5. Analyze

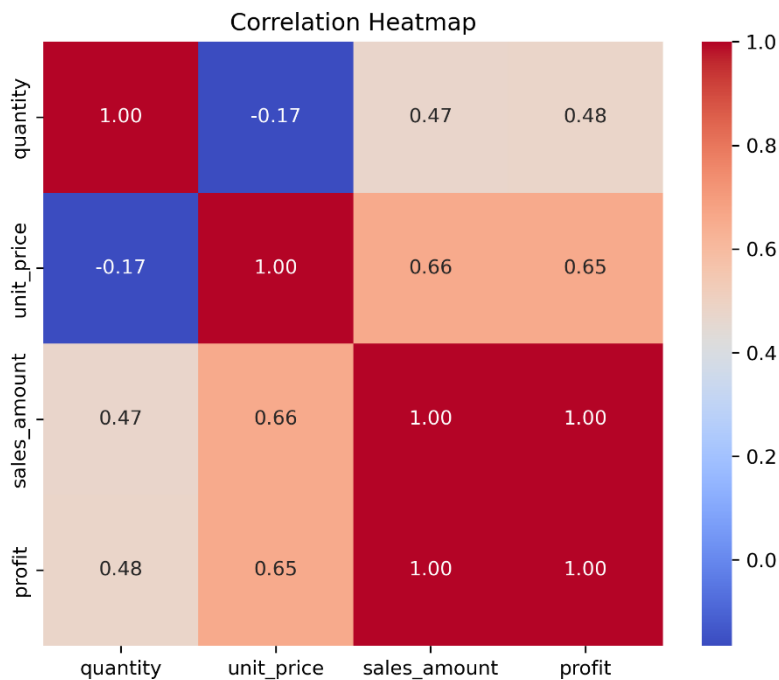
5.1 Descriptive Statistics



Metric	Quantity	Unit Price (₹)	Sales Amount (₹)	Profit (₹)
Mean	4.48	6,039.00	25,030.52	10,103.02
Median (50%)	4.5	5,249.00	16,743.50	7,641.50
Std Dev	2.76	4,553.83	24,582.52	9,766.51
Min	1	799.00	799.00	349.00
Max	9	14,999.00	104,993.00	42,693.00

Finding: The mean sales amount (₹25,030.52) sits well above the median (₹16,743.50), indicating a right-skewed distribution driven by a small number of high-value transactions.

5.2 Correlation Analysis



Variable Pair	Correlation
Sales Amount ↔ Profit	0.998
Unit Price ↔ Sales Amount	0.656
Quantity ↔ Sales Amount	0.474
Quantity ↔ Unit Price	-0.165

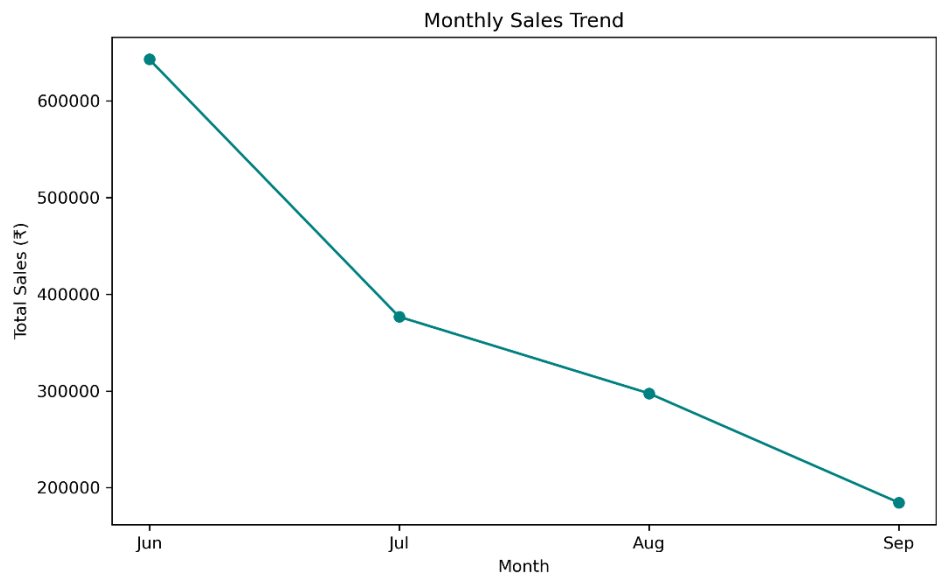
Finding: Sales Amount and Profit move in near-perfect lockstep (0.998), confirming profit scales directly with revenue. Unit Price is a notably stronger revenue driver (0.656) than Quantity (0.474) — **pricing and product mix matter more than order size** in this dataset

5.3 Outlier Detection

Sale ID	Product	Quantity	Sales Amount (₹)	IQR Upper Bound
5008	Gaming Chair	7	104,993	77,617.50
5011	Gaming Chair	6	89,994	77,617.50

Finding: Two Gaming Chair transactions exceed the IQR upper bound. Both were retained as legitimate bulk orders rather than removed as errors — a business validation step, not automatic deletion.

5.4 Monthly Trend



Month	Total Sales (₹)
June	643,016
July	376,741
August	297,530
September	184,544

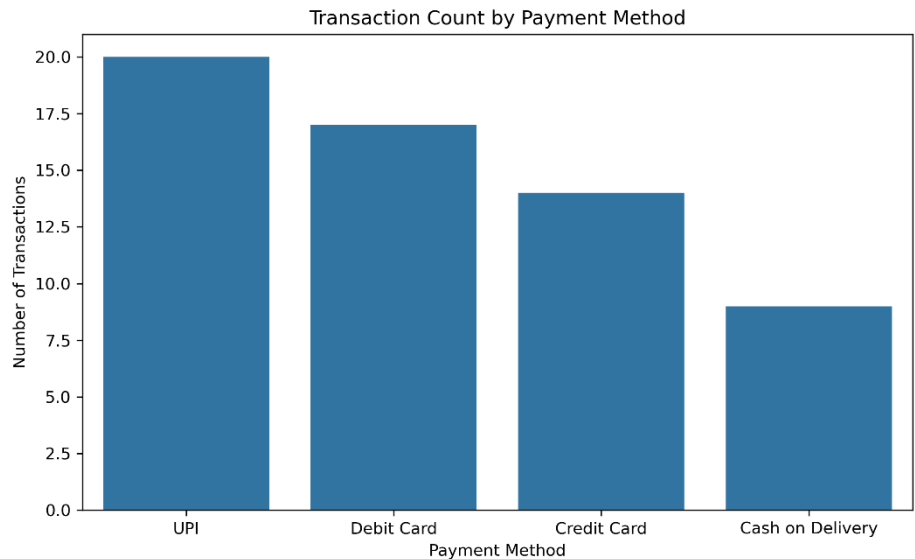
Finding: Revenue declined every single month, a drop of approximately **71%** from June to September — the single most urgent pattern in the dataset.

5.5 Regional & Category Breakdown

Region	Total Sales (₹)	Category	Total Sales (₹)	Share
South	414,840	Storage	509,940	34%
East	410,239	Furniture	329,978	22%
West	342,917	Electronics	254,046	17%
North	333,835	Wearables	208,981	14%
		Accessories	198,886	13%

Finding: South and East together account for over half of total revenue. Storage is the single largest category, contributing over a third of all sales.

5.6 Payment Method Performance



Payment Method	Total Sales (₹)
UPI	404,902
Debit Card	403,224
Cash on Delivery	398,254
Credit Card	295,451

Finding: Credit Card trails the other three methods by a consistent, meaningful margin (~₹100K+ gap from UPI).

6. Share

Power BI Dashboard Overview

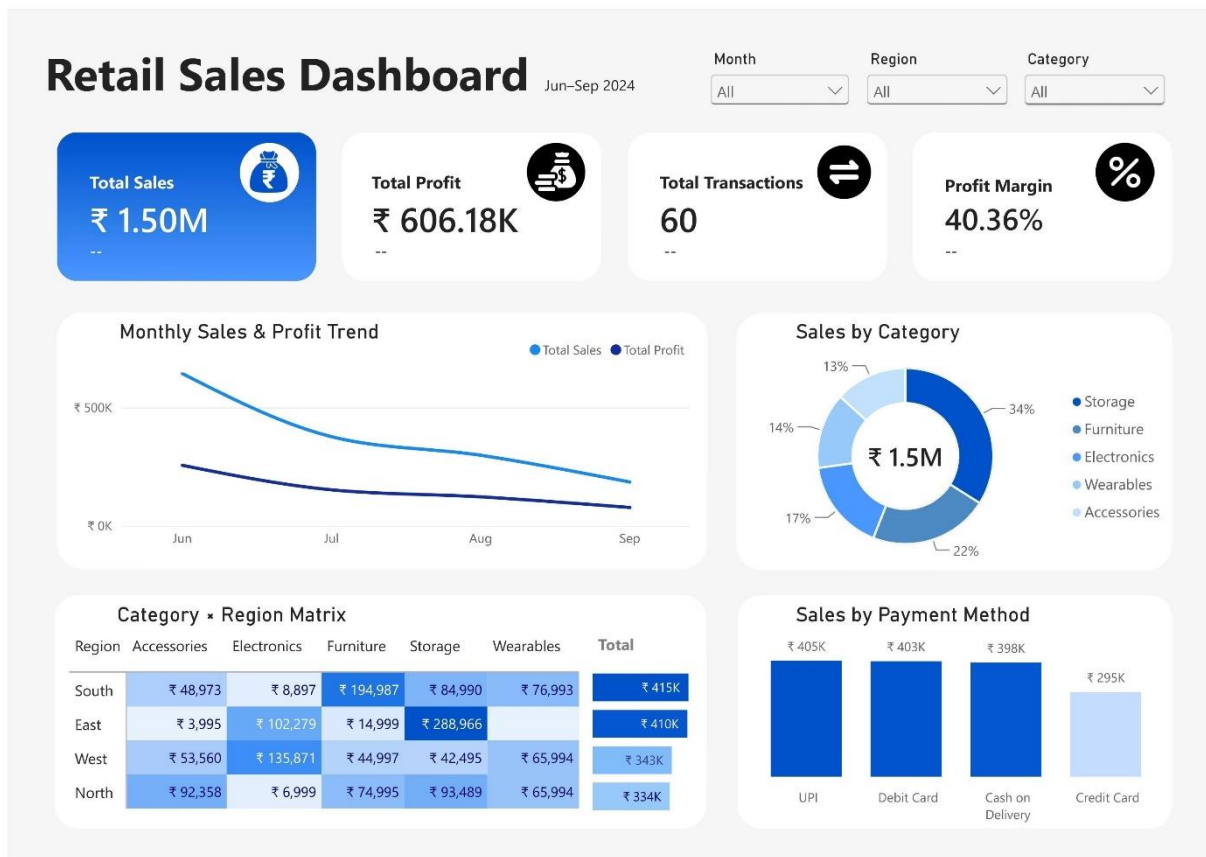
The final deliverable is an interactive Retail Sales Dashboard, built on the same cleaned dataset (sales_data_cleaned) used in the Python analysis above, ensuring both tools produce identical top-line figures.

KPI Cards

KPI	DAX Measure	Result
Total Sales	SUM(sales_data_cleaned[Sales Amount])	₹1.50M
Total Profit	SUM(sales_data_cleaned[Profit])	₹606.18K
Total Transactions	COUNTROWS(sales_data_cleaned)	60
Profit Margin	DIVIDE([Total Profit], [Total Sales Amount], 0)	40.36%

Core Visuals

Visual	Type	Fields
Monthly Sales & Profit Trend	Line Chart	X: Month, Y: Total Sales, Total Profit
Sales by Category	Donut Chart	Legend: Category, Values: Total Sales
Sales by Region	Bar Chart	Y: Region, X: Total Sales
Sales by Payment Method	Column Chart	X: Payment Method, Y: Total Sales
Category × Region Matrix	Matrix	Rows: Category, Columns: Region, Values: Total Sales



Finding: The mean sales amount (₹25,030.52) sits well above the median (₹16,743.50), indicating a right-skewed distribution driven by a small number of high-value transactions.

7. Act — Recommendations

Recommendation 1: Investigate and Reverse the Q3 Revenue Decline

Evidence: Monthly sales fell every month from June (₹643,016) to September (₹184,544) — a ~71% decline, while profit margin (40.36%) remained healthy, indicating a volume/demand problem rather than a cost problem.

Action: Cross-reference monthly transaction counts against marketing activity, stock availability, and seasonality using the dashboard's Month-over-Month measures to isolate whether the decline is driven by fewer transactions or smaller average order values.

Expected Reach/Timing: Immediate (Q4 planning cycle) — this is the single highest-priority action, as every other recommendation is secondary to stabilizing top-line revenue.

Recommendation 2: Concentrate Inventory and Marketing on Storage in South/East

Evidence: Storage is the top category (34% of revenue, ₹509,940) and South (₹414,840) / East (₹410,239) are the top two regions, together outperforming West and North significantly.

Action: Prioritize stock availability and targeted promotions for Storage products specifically in South and East before expanding investment into lower-performing region-category combinations.

Expected Reach/Timing: Next 4–6 weeks — low-risk, high-confidence action based on proven demand rather than speculative expansion.

Recommendation 3: Diagnose the Credit Card Payment Gap

Evidence: Credit Card generates ₹295,451 versus ₹398K–₹405K for the other three payment methods — a consistent ~27% gap too large to dismiss as random variation.

Action: Audit whether Credit Card transactions face structural friction (processing fees, verification steps, slower confirmation) versus UPI/Debit Card; a targeted UX or incentive fix could recover meaningful revenue share.

Expected Reach/Timing: Within one quarter — requires coordination with payment processing/checkout teams, so timeline is moderate rather than immediate.

8. Additional Data for Future Analysis

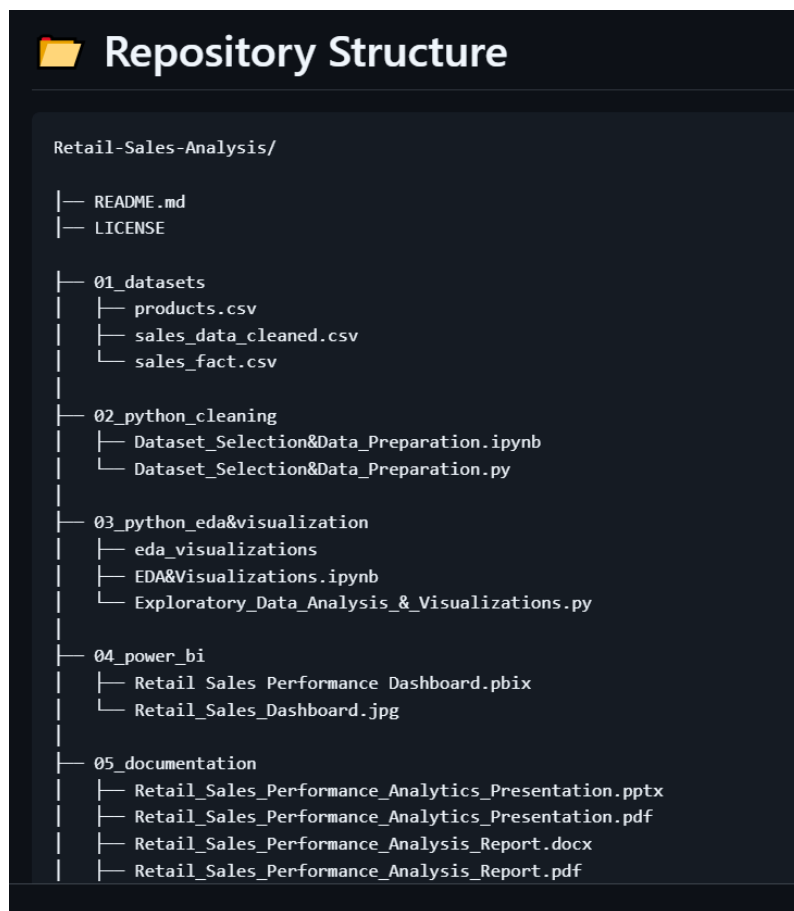
To strengthen future iterations of this analysis, the following data would be valuable:

- Customer demographic data (age, location, loyalty tier) to segment revenue drivers beyond region alone
- Marketing spend and campaign data to directly test correlation with the observed monthly decline
- Inventory/stock-level records to distinguish demand-driven decline from stock-availability constraints
- A longer historical time window (12+ months) to separate genuine trend from seasonal effects
- Customer acquisition vs. repeat-purchase data to understand whether the decline stems from fewer new customers or reduced repeat orders
- Competitor pricing or market data for external context on the Q3 slowdown
- Payment processing fee structure to directly test the Credit Card underperformance hypothesis

. Methodology & Reproducibility

Phase	Tool	Script / File	Output
Ask	—	This report, Section 2	Business task & stakeholder definition
Prepare	—	sales_fact.csv, products.csv	Raw data inventory & ROCCC assessment
Process	Python (Pandas)	Dataset_Selection_Data_Preparation.ipynb	sales_data_cleaned.csv (60 rows, 13 columns, 0 nulls, 0 duplicates)
Analyze	Python (Pandas, NumPy)	EDA_Visualizations.ipynb	Descriptive stats, correlation matrix, outlier detection, 8 visualizations
Share	Power BI Desktop	Retail Sales Dashboard (.pbix)	4 KPIs, 5 visuals, 3 slicers
Act	—	This report, Section 7	3 data-driven recommendations

GitHub Repository Link: [Retail-Sales-Analysis](#)



End of Case Study — Retail Sales Performance Analytics

InternNova Data Analytics Internship

Week 6 Assignment: Git, GitHub & Final Data Analytics Project

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